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An Offline, Automatic Mixing Method for Live Music, Incorporating Multiple Sources, Loudspeakers, and Room Effects

Abstract: A model of live performance is presented that includes simplified acoustic-environmental system models and enables the coupling behavior between multiple sources and receivers to be predicted. The model allows the mix at each location within the performance space to be evaluated as a function of the acoustic signals generated by the instruments and of the control parameters on the mixing desk, through which the instrument signals are sent before being reinforced using loudspeakers. For a set of listener locations, which includes both performers and members of the audience, ideal mixes are defined, and an optimization algorithm is developed that sets the control parameters on the mixing desk automatically, to deliver approximations of these ideal mixes to all listener locations simultaneously. The control parameters are constrained during the optimization to prevent the onset of acoustic feedback. The algorithm is examined, and we show that a targeted approach, which first sets the control parameters relating the vocal level, gives a better solution in a shorter time when compared to a brute-force approach.

The audio signal paths at a live, sound-reinforced music performance are modeled using the flow chart shown in Figure 1. The music is performed by a group of performers, each of whom plays an instrument. (The voice is considered an instrument for the purposes of this article.) Each instrument produces an audio signal. ("Audio" includes acoustic signals as well as electrical ones.) In some instances, such as with a drum kit, an instrument will produce multiple audio signals. For the purposes of this article, an instrument that generates multiple audio signals is treated as a number of individual instruments. Receivers are either human listeners (i.e, members of the audience or performers) or microphones. There is a signal path to the receiver directly from the instrument (the direct path indicated by the dashed lines and arrows on the left side of Figure 1), and a path via the mixing desk and sound-reinforcement system (the reinforced path indicated by the solid lines and arrows on the right side of Figure 1), the outputs of which are loudspeakers. The signals from instrument or loudspeaker to the receiver are acoustic signals, and are affected by the acoustic medium and the room. The signals from the mixing desk and input to the loudspeakers are electrical signals. The instrument audio signals enter the mixing desk via an acoustic signal path and a pressure transducer (i.e., a microphone) or an electrical direct input (DI) signal path. Once inside the mixing desk, the electrical signals can be transformed using signal-processing tools—for example, equalization. The acoustic signal that reaches the receiver is a linear superposition of all signal paths from all instruments. It is a mixture of musical signals, and in this article is used as the definition of a *mix*.

The mix experienced by the audience is called the front-of-house (FOH) mix. The mixes experienced by the performers are called monitor mixes. A uniform FOH mix is desirable at all audience locations, whereas each performer has a different ideal monitor mix. The sound-reinforcement system contains designated FOH and monitor loudspeakers. In large venues, the contribution of the direct signal path to the FOH mix becomes negligible, owing to the distances between the instruments and the audience. In smaller venues it can be very significant. The close proximity of the performers to their instruments means that the direct signal path always contributes to the monitor mixes, even in large venues. The mixes experienced by

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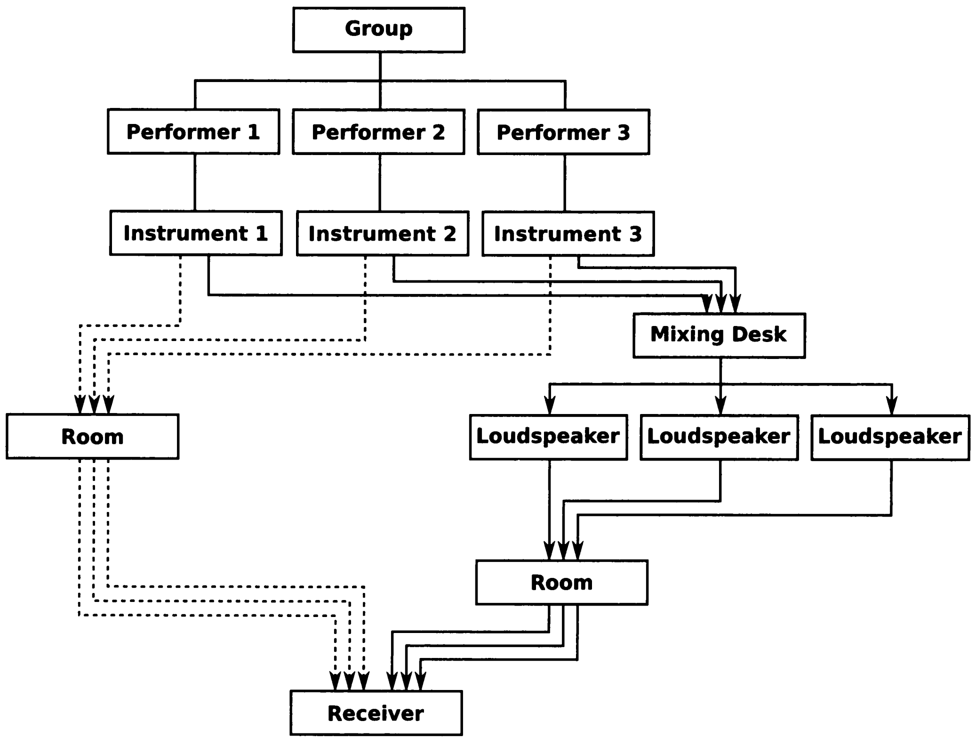
Figure 1. The setup and signal paths at a live musical performance. The group is composed of a number of performers, each of whom plays one or more instruments. The dashed arrows represent the direct signal path from instrument to receiver, and the solid arrows represent the reinforcement signal path from instrument to receiver via the mixing desk and sound reinforcement system.



the listeners are interdependent. It is not possible to make changes to one mix without affecting all others. This is referred to as *coupling* between mixes. In larger venues, the increased distances between receivers, instruments, and loudspeakers means that coupling between FOH and monitor mixes is minimal. In smaller venues, strong coupling can exist between all mixes. Mixes cannot be set independently, and a best fit to the requirements of all listeners must be sought.

It is the responsibility of the mixing engineer to ensure that the mix experienced by each listener is as desired. This is done by adjusting control parameters on the mixing desk, such as gain and equalization, to transform the electrical signals in the reinforced signal path. These changes can only be made while experiencing the mix at the mixing desk location (it is possible for the engineer to check multiple locations, but the controls cannot be adjusted while away from the desk unless a remote control is available). The engineer cannot experience the mixes at the performer locations, and so he or she can only infer the required changes in the mixes through communication with the performers. The changes that can be made to the control parameters are constrained. Two constraints present at all musical performances relate to the maximum allowable sound pressure level (SPL) and the onset of acoustic feedback. The former is more restrictive at venues in residential areas, which have a lower SPL limit. The latter restricts the transformations that can be applied to the electrical signals which have been input to the mixing desk via a microphone. It is problematic with instruments that have a low SPL; in popular music, this is most common with vocals.

The Engineer's Role as an Optimization Problem

The task with which the engineer is faced can be viewed as a constrained, multi-objective optimization problem. Each objective relates to the mix at a given receiver location and, because the mixes are coupled, they cannot be considered independently. The constraints relate to the maximum SPL and acoustic feedback, and the parameters within the optimization problem are the controls on the mixing desk. Three assumptions are made at this point.

1. The performance can be modeled accurately, i.e., if the instrument signals and mixing desk controls are known, it is possible to evaluate the mix at any location.
2. The mix at a given location is defined as the linear superposition of multiple acoustic signals at that location, and a particular mix can be described using a set of features extracted from the acoustic signals from which it is composed. No assumption is made at this stage with regard to the form of these features.
3. The set of features which describes the desired mix at each listener location, hereby referred to as the *target mix*, is available.

If these assumptions hold, then it is possible to formalize the optimization task undertaken by the engineer. The degree to which each objective is satisfied can be evaluated by comparing, at each listener location, the features of the mix with the features of the target mix. The total error is evaluated by combining the error measurements of all objectives. The set of parameters, i.e., the controls on the mixing desk, can then be sought which minimizes the error. This optimization will replicate the role of the engineer, and will enable the mixing desk controls to be set automatically to deliver the features of multiple target mixes to multiple listener locations during a live performance.

Automatic Mixing

The field of automatic mixing concerns algorithms or tools that automate, or assist in, aspects of musical production. Although there are diverse approaches to and uses of automatic mixing, the objective of all automatic mixing algorithms or tools is to set the control parameters on an audio signal-processing device such that the features of the processed signal are more desirable. The work in this article clearly falls into this category.

Automatic mixing tools can be classified as real-time or offline. Real-time automatic mixing tools continually evaluate the objective and provide the corresponding control data. Real-time automatic mixing tools are adaptive digital audio effects (A-DAFx), as defined by Verfaille, Zölzer, and Arfib (2006). Offline automatic mixing tools analyze a predefined audio signal (or group of audio signals) and derive a single, static set of control parameters. When applied to live music, offline tools can replace the sound check by setting a baseline from which the audio engineer can deviate in real time.

The first real-time automatic mixing tools were developed by Dugan (1975, 1989) and Julstrom and Tichy (1976) for conference applications. The objective of these tools was to prevent amplification of background noise and the onset of acoustic feedback. Dugan used a microphone placed in the conference space to gain an estimate of the background noise level. Noise gates were used on each microphone channel, and the threshold on each gate was set using the estimated background noise level. Julstrom used front- and rear-facing microphones at the location of each speaker (i.e., each conference participant). The direction of the sound source was estimated by comparing front and rear signals, and signals identified as coming from the rear (i.e., not the direction of the speaker) were attenuated.

Real-time automatic mixing tools for live music production have been presented by Gonzalez and Reiss (2007, 2008, 2009a, 2009b, 2010). Each tool is focussed on a specific aspect of production. The automatic gain normalizer (Gonzales and Reiss, 2008) sets the input gain on the mixing desk. The objective is to maximize the level to give good dynamic range while preventing distortion and clipping. The automatic panner (Gonzales and Reiss 2007; Gonzales and Reiss 2010) pans channels based on spectral content. The objective is to reduce masking by spatially separating instruments with similar frequency content. Additional subjective considerations are used to make the resultant panning more representative of a human mixing engineer—for example, low-frequency sources and vocals were not panned. Through subjective evaluation, it was shown that the panning settings obtained from the automatic panner were equally good to those set by an experienced engineer. The automatic fader and equalizer effects (Gonzales and Reiss 2009a, 2009b) set fader levels and a filter bank to ensure that the probable loudness of all instrument signals is equal. It deals with specific choices in relation to the second two assumptions listed earlier in the previous section. The feature used to describe the mix is the loudness of each instrument, and the target mix consists of instruments with equal loudness probability. When combined with the automatic gain normalizer and the automatic panner, the result is a general definition of a target mix which can be applied to any group of instruments. Though this generality is very useful if no knowledge of the music is available, it does not permit specification, for example by the artist, as to how the mixes should sound. In other words, musical criteria might rule out equal loudness of instruments. The work by Gonzalez and Reiss is targeted toward the FOH mixes at large venues. They assume that the FOH mix can be set independently from the direct signal paths and the monitor mixes, and that the sound system and room response has been equalized for all audience locations using sound-system-engineering methodologies. The reader is referred to a comprehensive text from McCarthy (2007) on sound-system engineering. These assumptions are valid for large venues, and so a model of the performance which accounts for the interaction between sources and receivers, as proposed in the introduction to this article, was not needed.

Offline automatic mixing tools for post production have been presented by Kolasinski (2008) and Barchiesi and Reiss (2009, 2010). In their work, the control parameters applied to audio effects on individual tracks within a multi-track mix were sought such that when the tracks were combined, the resultant mix had features matching a target mix. Kolasinski estimated the gain applied to each track, and used the spectral histogram of the mix as the feature that described it. This resulted in a non-unique solution, so a heuristic optimization algorithm was used. The problem of non-uniqueness was exacerbated when the number of tracks was increased. Barchiesi and Reiss sought the control parameters for equalization filters and delays applied to each track, as well as for gain controls. Instead of using higher-level features of the mix, the full time-domain signal in the mix was modeled as a linear combination of the individual tracks, which had been convolved with linear, time-invariant (LTI) impulse responses. The impulse responses were then extracted on a frame-by-frame basis using the least-squares method. If the equalization applied to the tracks was LTI (i.e., performed using an finite-impulse response filter), and the estimation order of the filter was at least as long as the equalization filter, then the exact mixing parameters could be found. For this reason their work was termed "reverse engineering of a mix." In their work, they also identified the nonlinear gain envelopes that had been applied to the tracks using a compressor. This was done by modeling the envelope across small frames of audio using a polynomial function. Because of the inherent nonlinear nature and variability in compressor implementation, it was not possible to relate the gain envelopes directly to compressor parameters.

An offline, automatic monitor mixing technique for live music production was presented by Terrell and Reiss (2009). In their model, the root-mean-square (RMS) SPL of each instrument at each performer location was evaluated by combining the RMS SPLs of the acoustic signals from the direct signal path and the reinforced signal path in anechoic conditions. The SPLs were combined independently of frequency, assuming incoherence between signals, and the mix was described using the relative SPL of each instrument. The authors subjectively defined target mixes and, using a gradient-descent method, they optimized the gain applied to each channel on the mixing desk to give a best fit to the target mixes at four performer locations. The importance of the different target mixes was weighted to reflect that the quality needed in a monitor mix can differ from one performer to the next. In this case, the order of importance was rated to be: vocalist, guitarist, bassist, then drummer. Constraints were placed to keep the feedback-loop gain below −3 dB, and to keep the RMS SPL of the monitor mixes between

95 dB and 105 dB at each receiver location. It was shown that the feedback constraint severely restricted the feasible solution space and resulted in the mix features at the drummer location being far from the target. It was also shown that the algorithm could be used to optimize the stage layout as well as the mixing desk controls. The interaction between acoustic signals generated by the same instrument, but emitted from different sound sources, was not included in the model, nor was the interaction with early room reflections. This was a significant simplification.

A final body of work which relates to automatic mixing is the re-parameterization of an audio-processing device, typically a standard audio effect, so that it can be controlled using higher-level, and in some cases perceptual, attributes. Reed (2000) presented a system that learns the equalization adjustments needed to effect a certain type of perceptual change. His system was trained by asking users to make specific perceptual changes to training audio. Once trained, the user could, for example, "make the song sound brighter" using a single control. A similar approach has been taken by Rafii and Pardo (2009) and Sabin and Pardo (2009a, 2009b), in the training of reverberation and equalization effects, respectively. Terrell, Reiss, and Sandler (2010) presented an algorithm that automatically set the attack, release, hold, and threshold parameters of a noise gate applied to a kick-drum recording containing bleed from secondary sources. The objective was to minimize the distortion of the kick drum while ensuring a predefined reduction in the bleed level. Once the parameters had been set, the gain control could be used to control the strength of the noise gate.

Direction Taken in this Article

Part of the motivation behind this work is to give artists more control over their live production. In order to do this, the features of the target mixes should relate to specific songs, as opposed to having general target-mix feature rules, as used by Gonzales and Reiss. Implementing specific, real-time target-mix features would necessitate a dynamic description of the target mix and a knowledge of the temporal position within the song. Although this is an interesting idea, it is beyond the scope of this work; we chose an offline approach instead. Offline approaches provide a static set of control parameters, and these are derived from a specific section of audio. This is analogous to the sound check, during which the engineer will set up good, general mixes, to which relatively minor adjustments are made during the performance. The work in this article is, therefore, focused on providing good, baseline mixes, equivalent to those produced during a sound check. An advantage of the automated approach is the ability to evaluate mixes at different locations simultaneously and to take into account both the coupling between mixes and the contribution of the direct signal path. A key contribution of this study is the development and inclusion of simplified acoustic-environmental-system models, able to predict the coupling behavior between multiple sources and receivers. The case study focuses on smaller venues, because the coupling effects are stronger than in larger ones.

The Live Performance Model

Instruments and loudspeakers produce acoustic signals and are collectively referred to as acoustic sources. Instruments can also produce electrical signals which are input directly to the mixing desk, but these are not included in this study. The definition of an instrument in this article includes sources of amplification that are specific to the instrument. For example, a guitar amplifier is part of the electric guitar instrument. We model acoustic sources as point sources from which the acoustic signal is radiated (the point lies within the geometry of the acoustic source). The radiation pattern of a real acoustic sources is a function of frequency and can be nonlinear.

Idealized models of loudspeaker radiation patterns have been presented in the literature. An example is a piston in an infinite baffle (Beranek, 1954). For non-theoretical applications, such as line-array modeling (Meyer, 1984), measured radiation patterns are used. The radiation pattern is represented by a set of

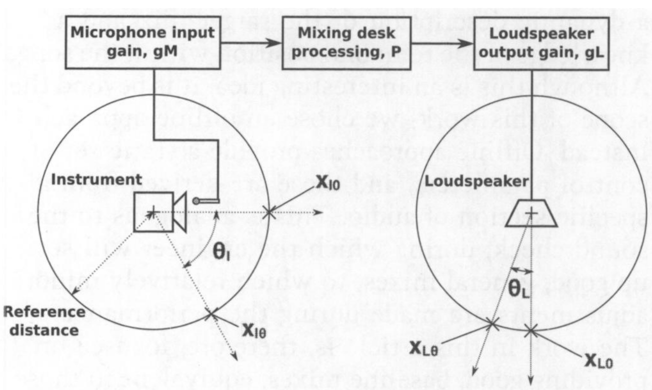


Figure 2. An illustration of the electrical signal path from instrument to loudspeaker. It is assumed that the microphone gain and loudspeaker amplifier gain are set such that the on-axis signal 1 m from the loudspeaker is equal to the on-axis signal 1 m from the instrument when the mixing desk controls are set, so that the signal entering the mixing desk is exactly equal to the signal leaving it.

impulse responses for a discrete set of radiation directions, measured at a reference distance beyond which the radiation pattern of the sound field can be treated as being uniform. Inside the reference distance, termed the *near-field*, the radiation pattern is not uniform owing to interactions between different parts of the acoustic source that generate an acoustic signal, and interactions between the acoustic signal and the particular geometry of the acoustic source. Radiation patterns can be measured accurately for loudspeakers, because a test signal can be used for the loudspeaker input. This is not the case with instruments, such as a voice or a drum kit, which do not have their own separate amplification system. Even if such radiation patterns were available for all instruments, difficulties would still arise, because the microphones used to transfer the acoustic signals to the mixing desk are in the instrument near-field, where the radiation pattern is not uniform.

For simplicity, we model the radiation patterns of acoustic sources and loudspeakers using two-dimensional polar broadband gain functions. We define x_{I0} as the on-axis, acoustic time-domain signal produced by an instrument at a distance of 1 m. The off-axis acoustic signal $x_{I\theta}$ is related to this by

$$x_{I\theta} = x_{I0} R_I, \tag{1}$$

where R_I is a polar broadband gain function. Figure 2 shows the signal path from an instrument to a loudspeaker. It shows the instrument, the microphone, the gain control on the microphone input g_M, the mixing desk which applies a linear transfer function P, the loudspeaker amplifier gain g_L, and the loudspeaker. Instrument acoustic signals are sent to the mixing desk via a designated microphone. Bleed from other instruments is omitted in our model, and we identify microphones by the instrument to which they are assigned (i.e., microphone 1 is assigned to instrument 1).

We assume that g_M and g_L are set to make the signal at the loudspeaker on-axis reference distance equal to the signal at the instrument on-axis reference distance when P is a unity broadband gain function. g_M and g_L are static controls, external to the mixing desk, and are not included as parameters in the automatic mixing algorithm. x_{L0} and $x_{L\theta}$ are acoustic signals at the loudspeaker on-axis and off-axis reference distances, respectively. In the frequency domain, where X is the Fourier transform of x,

$$X_{L\theta} = X_{L0} R_L = X_{I0} P_M R_L, \tag{2}$$

where R_L is the polar broadband gain function describing the loudspeaker radiation pattern.

We model the response of receivers using the polar broadband gain function, A_R, as with the acoustic sources; we model the listener responses as monaural. The source radiation and receiver-response polar-broadband gain functions are shown in Figure 3. The loudspeakers and amplifier radiation patterns are based on Meyer loudspeakers, and are approximated using Meyer's Mapp online software (Meyersound, 2011). Vocal and drum radiation patterns are modeled as omnidirectional, as is the listener response, and the microphone response is modeled as a perfect cardioid.

The path of the acoustic signal from an acoustic source to a receiver is described by a room impulse response. This encapsulates all room acoustic effects. There is much research on the modeling of room impulse responses (RIRs). Kuttruff's *Room Acoustics* (1979) is a comprehensive text on the subject. There are three main methods: analytical, geometrical, and finite-difference time domain (FDTD). Analytical methods solve the wave equation analytically, and are only applicable to very

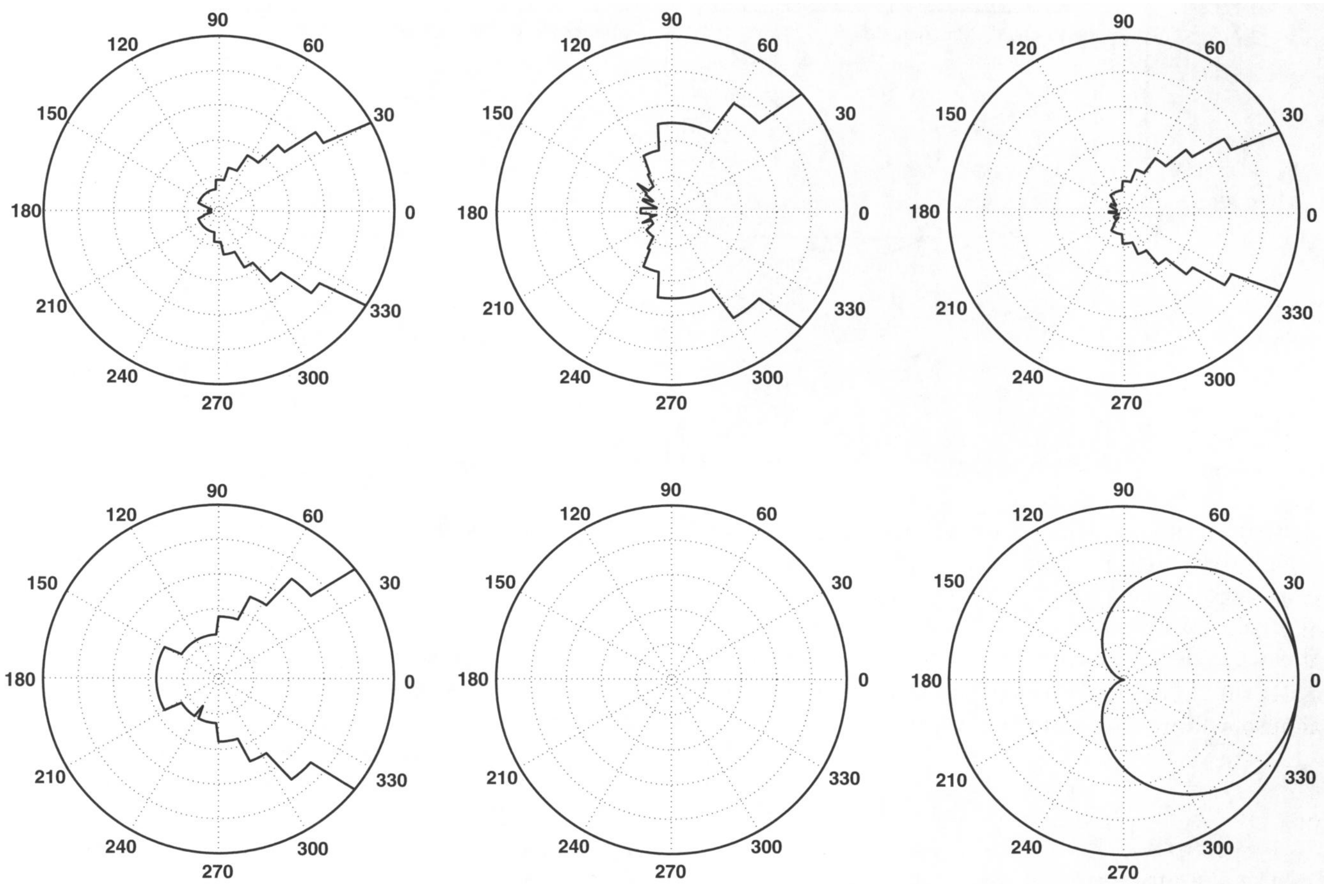
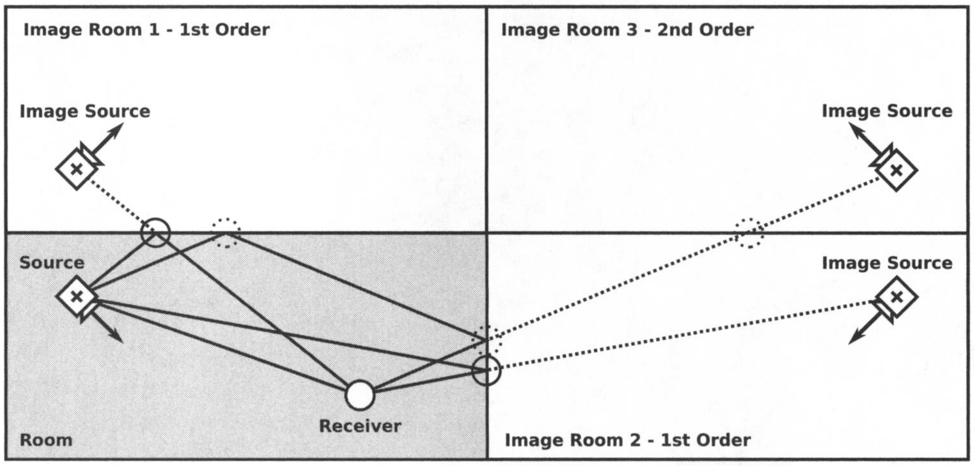


Figure 3. Source dispersion and receiver response patterns. (a) FOH loudspeaker, based on Meyer UPA-1P @ 1k Hz; (b) monitor loudspeaker, based on Meyer UM-1P @ 1k Hz; (c) guitar amplifier, based on Meyer UPA-1P @ 2k Hz; (d) bass amplifier, based on Meyer USW-1P @ 250 Hz; (e) omnidirectional source/receiver used to model the dispersion of the vocals, all components of the drum kit, and the listener response; and (f) cardioid pattern used to model the microphone response.

simple geometries. Geometrical methods simplify the description of a sound wave to that of a ray, by dividing a spherical sound wave emitted from a simple omnidirectional source into discrete quanta of sound energy. FDTD methods discretize the room into a large number of simple elements and numerically solve the wave equation for each element. We use a geometrical method called "the image source method" for the purposes of this article. Early publications of the image source method were presented by Allen and Berkley (1979) and Gibbs and Jones (1972). The image source identifies the static, virtual positions of reflected sources by plotting images of the room and source about the room boundaries. This is illustrated in Figure 4, in which three image rooms have been plotted. The path of a sound ray from each image source to the receiver can be traced. Each point where this path crosses a wall is a reflection point. The order of an image room reflects the number of times the sound ray is reflected before reaching the receiver. The attenuation, resulting from distance g_d and the time delay δt, is evaluated from the length of the path d, and the wall absorption is modeled using a broadband gain g_α.

A sound ray leaving a second-order acoustic source image will reach the receiver location after a time δt and will have a gain of $RA_R g_d g_{\alpha_1} g_{\alpha_2}$, relative to the on-axis acoustic source signal at the reference distance of 1 m, where g_{α_1} and g_{α_2} represent the absorption for two different reflection points (the subscripts I and L, which discriminate between

Figure 4. Diagram illustrating the image source method. The dashed lines emanating from the image sources are the reflected sound-ray paths. The solid lines show the path from the real source to the receiver, and illustrate how the reflected paths travel around the room. The solid and dashed circles highlight the wall crossings, i.e., reflection points, for first- and second-order image rooms, respectively. For simplicity, only a few reflections and image sources are shown. (There is actually one first-order reflection per room wall, one second-order reflection per first-order image room wall, and so on.) The room impulse is evaluated by combining the sound-ray paths from all image sources.



instruments and loudspeakers, have been removed for clarity). The complete RIR, denoted by h, is evaluated by combining the sound rays from all acoustic source images. In the frequency domain, the acoustic signal from instrument i at the receiver location r, being reinforced using N_L loudspeakers identified by the subscript l, is given by

$$X_{R_{ir}} = X_{I0_i} * \left(H_{I_{ir}} + \sum_{l=1}^{N_L} P_{M_{il}} H_{L_{rl}} \right). \qquad (3)$$

It is a combination of the direct signal path from instrument to receiver and the reinforced paths, which have traveled through the mixing desk and have been output by the loudspeakers.

This model enables the mix experienced at each receiver location to be described relative to the instrument on-axis acoustic signal. We have made three simplifying approximations in our model. First, polar broadband gain functions are used to model the radiation patterns of the sources and the responses of the receivers. This will introduce errors in the predicted mixes and will distort the predicted acoustic feedback gain. Second, the image-source method does not include wavelength and phase information, which means that diffraction, refraction, and interference effects are not included. Third, reflections are modeled as being specular, which means that all of the reflected energy is contained within a sound ray, and that the angle of reflection of the sound ray is equal to the angle of incidence. Real reflections are to some extent diffuse, which means that some of the energy is spread out rather than being contained in a ray. Specular reflections are perfectly correlated with the original acoustic signal, and will increase the effect of coloration and comb filtering. These approximations all relate to the evaluation of the RIR, so once the automatic mixing algorithm has been established, it will be simple to introduce more accurate RIRs that are taken from measurements or that are generated using more advanced room acoustic models.

Mix Features and the Target Mix

There is much work in the literature which seeks to describe the timbral similarity between audio stimuli (Grey, 1977, 1978; Grey and Gordon, 1978), but to our best knowledge, there is no work in the literature that seeks to identify such features to describe similarity between mixes. In order to find perceptually relevant features, a detailed subjective evaluation would be needed, and this is beyond of the scope of this article. The feature used in this article to describe a mix is the relative RMS of each instrument, which was also used by Terrell and Reiss (2009).

We extract mix features for a section of a song (for example, a chorus) in which all instruments are active. A typical chorus in a rock band is 16 bars which, at 120 beats per minute is around 30 seconds long. Convolving signals of this length with transfer functions at each iteration of an optimization algorithm would result in a very long solution time. In order to reduce the optimization time, the incoherent average of Fast Fourier Transform (FFT) windows is taken. The length of the windows is set equal to the length of the room impulse response, which in this article is 1 second at 44.1 kHz. The acoustic signal at the receiver location can be evaluated in the frequency domain using Equation 3. The RMS SPL, given by p_{rms}, can also be evaluated in the frequency domain using

$$p_{rms} = \frac{1}{N}\sum_{n=0}^{N-1} |X|, \tag{4}$$

where N is the number of points in the FFT.

The Objective Function

The objective function is the formalization of the engineer's role as an optimization problem. In this section we discuss the various components of the objective function: the evaluation of mix errors, the choice of control parameters, and the inclusion of a constraint to prevent the onset of acoustic feedback.

Mix Errors

The absolute RMS SPL of each instrument in a mix is evaluated using Equations 3 and 4 and is given by m_{A_r}. The relative RMS SPL given by m_{R_r} is used to describe the mix and is expressed as a difference in dB from the absolute RMS SPL of an arbitrarily chosen reference instrument in the mix. For example, a mix with absolute RMS SPLs of

$$m_{A_r} = \begin{bmatrix} 80 & 78 & 83 & 76 \end{bmatrix}^T \tag{5}$$

with instrument one chosen as the reference is described by

$$m_{R_r} = \begin{bmatrix} 0 & -2 & 3 & -4 \end{bmatrix}^T. \tag{6}$$

The second, third, and fourth instruments are 2 dB lower, 3 dB higher, and 4 dB lower, respectively, than the first instrument. If the target mix, m_{T_r}, is described by

$$m_{T_r} = \begin{bmatrix} 0 & -3 & 1 & -4 \end{bmatrix}^T, \tag{7}$$

then the error e_r in the mix at receiver location r is given by

$$e_r = ||m_{T_r} - m_{R_r}||. \tag{8}$$

The total error in all mixes, ϵ_T, is found by summing the mix errors at each location subject to a diagonal weighting matrix W, which allows priority to be assigned to the requirements of some listeners:

$$\epsilon_T = e^T W e. \tag{9}$$

The objective of the optimization algorithm is to minimize ϵ_T. This is done by adjusting the mixing desk parameters that control the transformations in the reinforced signal path. An individual transformation is applied to each instrument and loudspeaker pair. These transformations are identified by $P_{M_{il}}$, where i is the instrument index and l is the loudspeaker index.

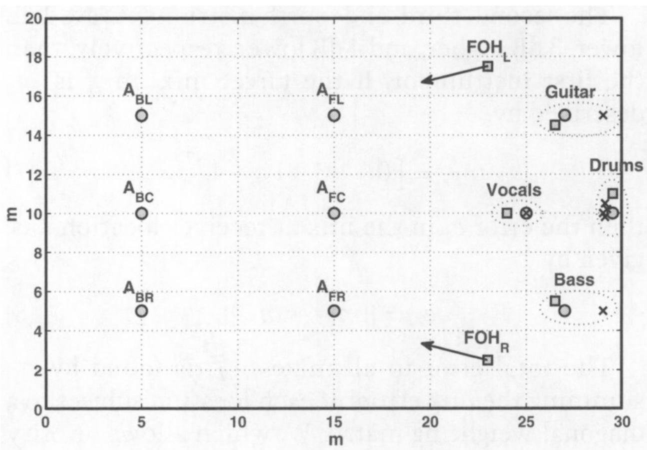
Control Parameters

The mixing control parameters used are broadband gains. This equates to sending a linearly scaled version of the instrument's acoustic signal through each loudspeaker. Real mixing desks also have equalization controls and additional audio effect processors, such as reverberation and compression, but these are not included here. Equation 3 can be rewritten as

$$X_{R_{ir}} = X_{I0_i} * \left(H_{I_{ir}} + \sum_{l=1}^{N_L} g_{C_{il}} H_{L_{rl}} \right), \tag{10}$$

where $g_{C_{il}}$ is the gain applied to the audio signal from instrument i before output via loudspeaker l. It is worth noting that Equation 3 can be use to implement any linear-audio transformation. This includes equalization and reverberation; dynamic audio effects, such as compressors, are nonlinear, so

Figure 5. Diagram showing the layout in the venue. Listeners are identified by ○, instruments by ×, and loudspeakers by □. Audience locations face the performers and are labeled using the convention A_{XY}, where x is: B for back or F for front, and y is: L, C, or R for left, center, or right, respectively. The performers, their instruments, and monitor loudspeakers are grouped and labeled. For guitar and bass, the instrument location is the amplifier; for the vocals, the instrument and performer locations coincide. Performers and instruments face the audience, and each monitor loudspeaker faces the performer to whom it is assigned. The orientation of the FOH loudspeakers are identified by arrows.



the model would have to be adapted if they were to be included.

Acoustic Feedback Constraint

van Waterschoot and Moonen's (2011) review of acoustic feedback control contains a detailed explanation of acoustic feedback. Acoustic feedback results from the coupling between loudspeakers and microphones, which can be described by a feedback-loop transfer function, F, which is given by

$$F = \sum_{m=1}^{N_M} \sum_{l=1}^{N_L} g_{C_{ml}} H_{L_{ml}}, \tag{11}$$

where m is the microphone index. Acoustic feedback will occur if the feedback-loop transfer function is unstable. The Nyquist (1932) stability criterion states that a closed-loop system is unstable if the loop gain is greater than or equal to unity, or if the loop phase is a multiple of a complete cycle at any frequency. We constrain the mixing controls to fulfill these criteria for the vocal microphone only (experience shows that vocal microphones are the most likely cause of acoustic feedback). During a performance, it is possible that the feedback transfer function can change—for example, if the vocalist changes position on stage with the microphone in hand. In order to account for this, a buffer of 3 dB is used, so the feedback constraint is given by $\hat{f} < -3$ dB, where $\hat{f}$ is the maximum magnitude of F. The review paper (van Waterschoot and Moonen, 2011) also describes state-of-the-art feedback control techniques. Many of these are online methods which use additional signal processing of the microphone signals to disrupt coupling between the loudspeakers and microphones to satisfy the Nyquist stability criteria. Such techniques are not considered here.

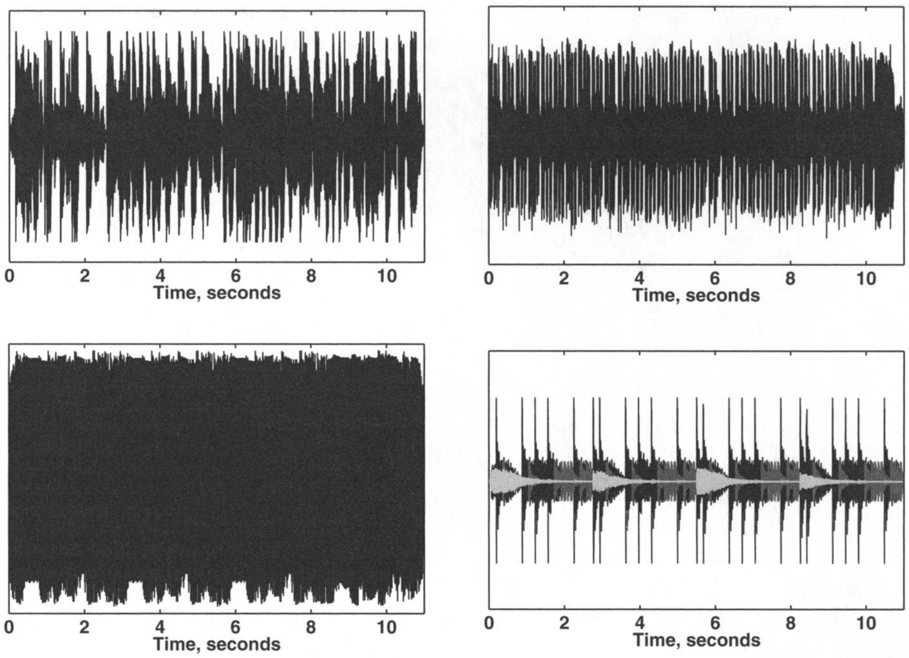
Case Study

We used a case study to demonstrate the proposed algorithm. It is based around a performance by a band consisting of four performers. Their instruments are: vocals, electric guitar, electric bass, and a drum kit, which is composed of a kick drum, a snare drum, hi-hats, and a cymbal. The performance takes place in a rectangular venue and the FOH mix is evaluated at six audience locations. The sound reinforcement system consists of two FOH loudspeakers and four monitor loudspeakers. The performance setup is shown in Figure 5. The wall behind the performers was modeled as having an absorption coefficient of 0.7, and the other walls with an absorption coefficient of 0.4. The 0.7 represents additional acoustic damping, such as heavy curtains commonly used at the back of the stage.

The audio for the case study is taken from the chorus of a song by an unsigned band. The audio for each instrument is plotted in Figure 6. For brevity, the signals from all parts of the drum kit are plotted together. The peak and RMS SPL of each signal at a reference distance of 1 m is contained in the figure caption. Target mixes were produced by the authors using a digital audio workstation (DAW) for each performer and for the audience. The relative RMS SPLs extracted from these mixes are

$$m_T = \begin{bmatrix} 0 & -4.2 & 2.6 & 0.4 & 0.5 & -18.2 & -14 \\ 0 & 1.7 & 5.2 & 2.7 & 2.6 & -16.2 & -11.9 \\ 0 & -3.7 & 8.4 & 8.5 & 6.8 & -16.2 & -11.9 \\ 0 & -2.3 & 7.7 & 6.5 & 6.1 & -14.6 & -12.6 \\ 0 & -0.4 & 6.7 & 2.7 & 2.8 & -18.6 & -11.8 \end{bmatrix}, \tag{12}$$

Figure 6. The on-axis acoustic instrument signals, (a)–(d) corresponds to vocals, guitar, bass, and the combined drums. Peak levels are 72, 90, 92, 96, 84, 65, and 80 dB SPL and RMS levels are 61, 77, 85, 81, 72, 45, and 61 dB SPL for vocals, guitar, bass, kick drum, snare drum, hi-hats, and cymbal, respectively.

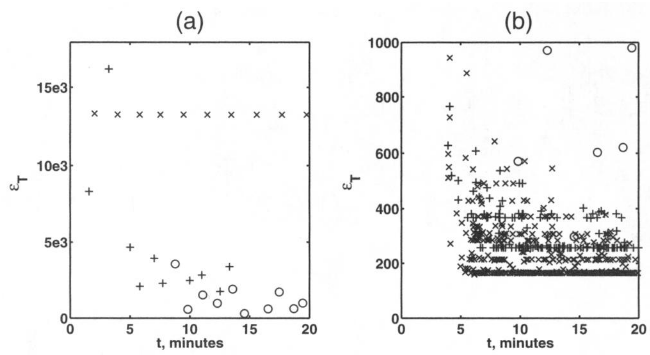


where rows 1 to 5 correspond to the vocalist, the guitarist, the bassist, the drummer, and the audience, respectively, and columns 1 to 7 correspond to vocals, guitar, bass, kick drum, snare drum, hi-hats, and cymbal, respectively. This ordering convention is maintained throughout this article. There are seven instruments and six loudspeakers, which gives 42 control parameters in total.

We constrained the control parameters relating to the FOH loudspeakers so that all instruments are panned to the center. Our model of the listeners does include panning perception, and this precaution prevents mixes with extreme or undesirable panning settings. This leaves 35 independent control parameters. The control parameters are optimized using a combination of a gradient-descent search method and a genetic algorithm, implemented using the Matlab *fmincon* and *ga* functions, respectively. The listener's requirements are weighted such that those of the vocalist and the audience are twice as important as those of the other performers.

Initial attempts to directly optimize the control parameters proved unsuccessful. Two causes are cited. The first is linear dependency between the control parameters. This can be understood by considering a situation where the SPL of instrument 2 must be increased relative to instrument 1. This can be achieved by either increasing the gain of instrument 2 or by decreasing the gain of instrument 1, and this can be done using any combination of available loudspeakers. This results in a solution space with many local minima, which causes problems for gradient-descent search methods. Figure 7a shows a plot of the residual against time when the gradient-descent search method is used. The solutions use a varying number of search iterations and are identified by a "×." It can be seen that the solution quickly becomes stuck in a local minimum and that, regardless of the number of iterations, no improvement is seen. The second problem is the feedback constraint. Terrell and Reiss (2009) demonstrated that the feedback constraint severely restricts the feasible solution space. Gradient-descent methods can become stuck on a constraint boundary, and genetic algorithms are slowed significantly because many more function evaluations are required at each generation to ensure that all members of the subsequent population are

Figure 7. The residual of the error function plotted against solution time. Part (a) shows results for the direct approaches in which: × uses the gradient-descent search, + uses the genetic algorithm, and ○ uses the genetic algorithm followed by the gradient-descent search. Part (b) shows results for the targeted approach in which + are members of cluster 1 and × are members of cluster 2. The direct approach using the combined optimization algorithms are also plotted and are identified by ○.



feasible. Figure 7a also shows the residual against time for the genetic algorithm, and a combination of the genetic algorithm and gradient-descent search method, identified by + and ○, respectively. A combination of the algorithms finds an improved solution, but the time required to find it is increased significantly, and it is unclear whether it represents a global minimum.

A more targeted, alternative approach is suggested. The genetic algorithm clearly shows better performance because it can deal with local minima; however, solving the entire optimization problem in one pass requires a significant optimization time. The factor affecting the genetic algorithm optimization time is the feedback constraint. The feedback constraint is a function of the vocal control parameters only (we have assumed that the vocal microphone is the most likely cause of acoustic feedback). By separating the optimization into two parts, the first of which sets the vocal control parameters only, the dimensionality of the constrained optimization problem is reduced, enabling a more thorough search of the solution space in a shorter time. Once the vocal control parameters have been set, the control parameters of the other instruments can be found quickly, as they are no longer constrained. Separating the optimization in this way also reduces the effect of linear dependency, because the control parameters of the remaining instruments are set relative to a fixed vocal level.

An objective function is required to set the vocal control parameters. It would be possible to use the existing objective function, but this would seek to deliver features of the target mixes as opposed to maximizing the vocal level before the onset of acoustic feedback. There are a number of ways that this maximizing could be done, and a full study is not considered here. The chosen objective is to make the vocal RMS SPL at each listener location higher than required with respect to all other instruments in the mix. The vocal error e_{v_r} is defined by

$$e_{v_r} = ||MAX(0, m_{R_{ir}} - m_{T_{ir}})||, \qquad (13)$$

where r is the listener location, i is the instrument index, and the vocal is the reference instrument used to describe the relative RMS SPL (see Equation 6). The total vocal error, ϵ_{vT}, is evaluated using the weighting matrix, as for ϵ_T in Equation 9. For each run of the targeted algorithm, this function is minimized, followed by the sequential minimization of the mix error for each instrument in turn, relative to the optimized vocal level. At each stage a combination of the genetic algorithm followed by the gradient-descent search method is used. To determine the required population size and number of iterations required to obtain a stable-solution genetic algorithm, population sizes from 5 to 50, followed by gradient-descent search method iterations of 5 to 50, are obtained. For each combination of algorithm parameters, ten solutions are obtained to account for the quasi-random nature of the genetic algorithm. The residuals are plotted against time in Figures 7b).

It is clear from Figure 7 that the targeted approach far outperforms the direct approach. There is a minimum in the solution of $\epsilon_T = 160$ and, although it cannot be proved definitively, this appears to be the global minimum. The consistency of the solutions is investigated using a k-means clustering method. This is an established machine-learning technique, further details of which can be found in Witten and Frank (2005). The number of clusters was estimated using Ward's (1963) method, which clearly indicated two clusters. In Figure 7b, solutions belonging to clusters 1 and 2 are indicated by + and ×, respectively. The correlation of all solutions with the centroid of the cluster to which they belong is very high. It is over 95 percent for all solutions for which the genetic algorithm population size is ten or more.

Table 1. Control Parameters (Gains) and Residual Errors for the Optimal Solution in Each Identified Cluster

			Vocals	*Guitar*	*Bass*	*Kick*	*Snare*	*Hi-Hat*	*Cymbal*	e_R
Cluster 1	Gain	FOH L	15.5	−2.8	−4.0	−8.2	1.9	12.5	5.4	
		FOH R	15.5	−2.8	−4.0	−8.2	1.9	12.5	5.4	
		Mon V	17.6	−2.7	−4.9	−4.6	4.3	15.0	5.1	
		Mon G	3.9	−24.9	−16.5	$-\infty$	−8.6	4.1	−6.1	
		Mon B	4.0	−16.3	−13.2	−12.0	−1.5	4.3	−5.9	
		Mon D	−72.9	−21.6	−15.8	$-\infty$	−7.7	−2.8	$-\infty$	
	Error	Vocalist	0	−0.6	0	0.4	0.4	0	0.4	0.9
		Guitarist	0	−3.3	0.1	−1.7	0.1	0	0	3.7
		Bassist	0	−0.3	−3.2	0.2	0.2	0	0	3.3
		Drummer	0	−0.3	0.2	−11.7	−2.2	0	−7.7	14.2
		Aud FL	0	−0.3	−0.2	−0.1	0.6	0	0	0.7
		Aud FC	0	0.3	1.4	1.1	0.3	0	0.1	1.8
		Aud FR	0	0.9	−0.8	−0.4	0.3	−0.1	0.1	1.3
		Aud BL	0	−0.4	−1.3	−1.4	−1.1	0.1	−0.2	2.3
		Aud BC	0	−0.1	−0.3	0	0.5	0	0.2	0.6
		Aud BR	0	−0.2	0.1	−1.3	−1.1	0.1	−0.1	1.7
		ϵ_T								**256.3**
Cluster 2	Gain	FOH L	15.8	−2.4	−4.0	−7.8	3.4	12.9	5.6	
		FOH R	15.8	−2.4	−4.0	−7.8	3.4	12.9	5.6	
		Mon V	8.9	−12.2	−14.6	−28.8	−6.2	6.4	−3.5	
		Mon G	4.2	−24.1	−15.9	−48.5	−7.5	4.0	−6.1	
		Mon B	5.5	−15.0	−11.7	−9.9	−0.5	5.2	−4.7	
		Mon D	5.5	−13.6	−11.2	$-\infty$	$-\infty$	5.4	$-\infty$	
	Error	Vocalist	0	0	0	−0.2	0	0	0.1	0.2
		Guitarist	0	−3.9	0.1	0	0	0	0	3.9
		Bassist	0	0	−2.8	0.2	0.1	0	0	2.9
		Drummer	0	0.1	−0.1	−9.2	−0.2	0	−4.9	10.4
		Aud FL	0	−0.4	−0.2	−0.3	0.4	−0.1	0	0.6
		Aud FC	0	0.2	1.5	0.9	0	0	0.1	1.8
		Aud FR	0	0.8	−0.7	−0.6	0.4	−0.1	0.2	1.3
		Aude BL	0	−0.4	−1.2	−1.5	−0.8	0.1	−0.1	2.1
		Aud BC	0	−0.3	−0.4	0.5	0.5	−0.1	0.1	0.8
		Aud BR	0	−0.1	0	−1.5	−0.7	0.1	−0.2	1.7
		ϵ_T								**158.8**

The gain section shows the gain in dB applied to each instrument in the indirect path via each loudspeaker. The error section shows the relative error of each instrument, the combined error at each listener location, e_R, and the total error for all listener locations combined, ϵ_T.

The control parameters and residual errors corresponding to the best solution in each cluster are shown in Table 1 (discussed subsequently). The gain controls generally show significant vocal amplification. There is minimal amplification (via microphones and loudspeakers) of guitar, bass, and kick drum, which demonstrates the contribution of the direct signal path to the mixes. Both clusters have small residuals at the audience and vocalist locations and large residuals at the other performer locations, in particular the drummer location. This is attributed to the weightings used and the increased

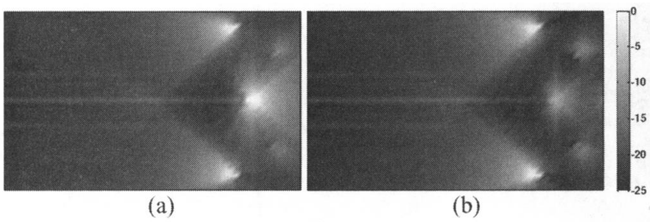


Figure 8. Intensity plots showing the relative SPL level of the vocal acoustic signal for the two solution clusters identified. Parts (a) and (b) correspond to clusters 1 and 2, respectively. In cluster 1, the vocal monitor is much louder (whiter), and its radiation pattern encompasses the drummer's location.

contribution of the direct sound of these performers' own instruments at their respective locations. This can be seen by inspecting the individual instrument errors. The largest errors for the guitarist, bassist, and drummer correspond to their own instruments, the levels of which are substantially higher than required.

The main differences between the two clusters is the vocal gain in the indirect path to the monitor loudspeakers. Cluster 2 shows a relatively even spread of vocal gain across the monitor loudspeakers, whereas cluster 1 shows a high concentration of vocal gain output via monitor loudspeaker 1 (the vocalist's monitor) and minimal vocal gain output via monitor loudspeaker 4 (the drummer's monitor). Figure 8a and Figure 8b are intensity plots of the RMS SPL of the vocal signals throughout the room for clusters 1 and 2, respectively. The scale is in dB and is relative to the peak RMS SPL of the vocal signal in the cluster 1 solution. The RMS SPL at the vocalist location is around 8 dB higher in cluster 1 compared with cluster 2. This is also true for all other instruments in the mix. Figure 8a shows that the acoustic signal from the vocal monitor loudspeaker provides substantial vocal SPL to the drummer location to compensate for the minimal vocal SPL from the drum monitor. It is debatable which of these solutions is best. The cluster 2 solution has a slightly lower residual and gives a more even SPL on stage. The cluster 1 solution indicates that it may be possible to remove the drum monitor and still obtain a reasonable mix at the drummer's location. Either solution is acceptable and can be found reliably using the targeted approach. The difference between the two solutions arises from the vocal gain settings, which are optimized in the first stage of the algorithm. It would be possible to add further constraints that favor one solution to the objective function, or to increase the population size or number of iterations to ensure that the lower residual corresponding to cluster 2 was found. The objective was to find a good approximation to the global minima in an acceptable solution time, and this has been accomplished, so these advancements are not considered at this stage.

We examined coupling between the mixes by setting the audience and performer mixes independently using only FOH and monitor loudspeakers, respectively. The control parameters for each case were then combined to produce the uncoupled mix. We compared the independent and uncoupled mixes to the coupled mix corresponding to cluster 2. The performance model used was identical to that used in the preceding case study, and we used the two-stage targeted algorithm to minimize the objective function with a genetic algorithm population size of ten and a gradient-descent method with ten iterations.

Table 1 shows the mix error e_R at each listener location, the total error ϵ_T, and the peak feedback transfer function $\hat{f}$ for the independent FOH and monitor mixes, the uncoupled mix, and the coupled mix. Differences between the mix errors of the independent and uncoupled mixes indicate the level of coupling (if there was no coupling in the model, the mixes in these two cases would be indentical).

The differences for the monitor mixes are minimal, except for the vocalist. The differences for the FOH mixes are slightly larger, with a maximum change of 1.5 dB at A_{BR}. This occurs because the audience is relatively closer to the monitor loudspeakers than the performers are to the FOH loudspeakers, when compared to the main source of sound reinforcement (FOH and monitor loudspeakers, for audience and performers, respectively). The SPL of an acoustic source decreases with the distance from the source, so the contribution of the monitor loudspeakers to the FOH mixes is greater than the contribution of the FOH loudspeakers to the monitor mixes. This effect is more pronounced for members of the audience at the back of the venue, and this is illustrated in the larger differences between FOH- and uncoupled-mix errors, as shown in Table 2. The vocalist is closer to the FOH loudspeakers than are the other performers, so the vocalist mix is affected by the FOH loudspeaker

Table 2. **A Comparison of the Residual in the Error Function and the Feedback-Loop Gain at Each Listener Location when the FOH and Monitor Mixes Are Treated as Being Coupled**

Mix	V	G	B	D	A_{FL}	A_{FC}	A_{FR}	A_{BL}	A_{BC}	A_{BR}	ϵ_T	$\hat{f}$
FOH	–	–	–	–	0.5	1.2	1.1	1.6	0.4	1	13.7	−8.9
Monitor	0.5	2.7	2	7.6	–	–	–	–	–	–	69.1	−3.0
Uncoupled	1.0	2.6	1.9	7.6	0.6	1.9	1.5	2.4	1.3	2.5	108.5	−1.3
Coupled	0.2	3.9	2.9	10.4	0.6	1.8	1.3	2.1	0.8	1.7	158.8	−3.0

signals. The performer mixes also have a larger contribution from the direct signal path, which is independent of the sound reinforcement system.

The coupled mix errors are smaller than the uncoupled mix errors for the vocalist and the audience, but are larger for the other performers, in particular the drummer. The FOH mixes contain a significant contribution from the monitor loudspeaker signals, so improved mixes are obtained by considering FOH and monitor loudspeakers simultaneously. This is also true for the vocalist, as discussed previously. The large differences in the guitarist, bassist, and drummer errors give an overall residual which is lower in the uncoupled case. Inspection of Table 1 shows that $\hat{f}$ is −1.3 dB for the uncoupled case, however, which is above the maximum allowable value. The difference between the guitarist, bassist, and drummer mixes lie mainly in the level of the vocals, which is higher in the uncoupled case but has been obtained by breaching the feedback constraint. Table 2 shows that for the independent FOH and monitor mixes $\hat{f}$ is −8.9 dB and −3.0 dB, respectively. Setting the monitor mix in isolation pushes the feedback constraint to the limit. Any further amplification of the vocals via the FOH loudspeakers breaches the feedback constraint.

The coupling between mixes can be considered in two ways. The first is simply the contribution of both sets (FOH and monitor) of loudspeakers on the mix at a given location. we see that the FOH and vocalist mixes are affected by both sets and that the monitor mixes of the other performers are affected minimally by the FOH loudspeakers. The result of this is improved mixes at FOH and vocalist locations when coupling is taken into account. The second way is the effect of the feedback constraint, which must be considered for the whole sound reinforcement system. The gain that can be applied to the vocal signal can be viewed as a resource that must be shared between FOH and monitor mixes. If set independently, it is likely that one set of mixes will use up all of this resource, leaving none for the other. This was shown to happen with the monitor mixes, which are clearly very sensitive to the feedback constraint.

Discussion

The case study shows that it is possible to deliver properties of multiple target mixes to multiple locations within a venue. This process approximates the role of the engineer and includes coupling between mixes and the contribution of direct sound. Both of these effects have been shown to be significant. In the examples presented, it was not possible to find an exact match to the target mixes, although the RMS mix errors for the audience and the vocalist were less than 1 dB. Listening tests would be needed to determine whether these differences are perceptible. The mixes cannot be set exactly, owing to the coupling between them and to the feedback constraint, which has a more restrictive effect on the monitor mixes. we also showed that setting the mixes independently results in a breach of the feedback constraint.

The targeted optimization algorithm gives a good approximation to the global minimum in a reasonable time. This approach splits the optimization into two parts, the first of which sets the vocal control parameters only. This is similar to a process known as "ringing out the monitors," in which equalization is applied to the loudspeaker inputs or microphone outputs in order to flatten the feedback-loop transfer function. The equalization allows more vocal gain to be applied before the onset of acoustic feedback. It can be applied to FOH or monitor loudspeakers, but is named after monitor loudspeakers, for which it is more prominent, as illustrated in the case study. The vocal level optimization presented in this article sets gain controls rather than an equalizer, but could be extended to include equalization. The advantage of this approach is that more-sophisticated requirements can be incorporated; for example, it is possible to define locations where the vocal SPL should be maximized instead of generally increasing the allowable vocal gain.

The model of live performance that we developed enabled the mix to be expressed as a function of the instrument acoustic signals and the RIRs between the acoustic sources and the receiver. Instrument acoustic signals were defined on axis at a reference distance of 1 m, and the electrical part of the indirect signal path was normalized such that if no processing is applied in this section of the indirect signal path, the loudspeaker on-axis acoustic signal is the same as the instrument acoustic signal. The RIR was modeled using the image source method, and the source radiation and receiver response patterns were modeled using polar broadband gain functions. Most of these modeling decisions are simplifications used to develop the automatic mixing algorithm, and can be advanced using established modeling techniques. The RIRs, for example, can be evaluated using FDTD methods, or they can be measured. In measuring the RIRs, it is also possible to take into account the source radiation and the receiver response patterns by measuring the RIRs with the performance equipment rather than specific test equipment. Binaural responses can also be evaluated using an ambisonic microphone or a dummy head. A part of the model which requires further research is the modeling of the acoustic instruments, e.g., vocals and drums for the case study shown in this article. It is not possible to measure the radiation pattern of an acoustic instrument, because the excitation signal cannot be defined and the response is subject to much variability. Incidentally, this will only impact the model if the contribution of the direct signal path is significant, so, for the vocal signal, it might not be a problem.

We describe mixes using the relative RMS SPL of each instrument. This feature represents the balance between the instruments, and informal evaluation showed that the mixes obtained at the audience locations (where the residual errors were minimal) were good matches to the target. There are many potential features that can be used, and the music information retrieval community is a good resource for such features. A formal evaluation of mix features, in particular those that can be used to define similarity, is needed, and is a topic of future work. It is intuitive that a similarity measure should include features representing the spectrum and the absolute SPL of the instrument signals. The former would require automatic equalization, which would be relatively simple to implement. The inclusion of the absolute SPL is more complex. If one listens to the same mix of a song at a very low SPL and then a very high SPL, is it the same mix? The answer is no, because the perceived loudness of each instrument in the mix is a function of its spectrum and absolute SPL. This implies that relative loudness may be a better measure of the balance of instruments in a mix.

The target mixes used in the case study were produced using a DAW. Whether this is a suitable way to define them is questionable, as they are not representative of live mixes. When considering the FOH mixes, this approach makes it easy for an artist to define how they want to sound, because even a simple home recording could be used. For monitor mixes this approach seems rather arbitrary, as it is very difficult to know how a monitor mix feels unless you are actually playing. Monitor mixes are generally at a very high SPL, and so the inclusion of an absolute SPL feature, as discussed earlier, is even more important than for the FOH mixes. It would likely be more useful to extract target monitor mixes from rehearsals or prior performances at which there is sufficient time in which to optimize them manually. Another problem with the mix definitions is their rigidity, that is, their precise definitions of low-level instrument features. This makes sense with FOH mixes because the objective is clearer, (e.g., "make the music sound like this...") but the objective for the monitor mixes is to enable the performers to perform to the best of their ability. We have shown that the precise target mixes cannot be delivered, and it is possible that using them may be unhelpful. The resultant mixes may be suboptimal if the optimization algorithm is (for example) trying to provide a specific guitar SPL relative to the vocals at the drummer location, when in fact the drummer is happy so long as the guitar is not drowning out the bass and the vocals. We suggest that the requirements of performers with respect to monitor mixes may be better described by higher-level features such as audibility and masking, particularly as it is unlikely that the prescriptive mixes will be obtainable.

Conclusion

A way to do automatic mixing for live musical production has been presented. The intended use is not as a fully automated system, but rather as a replacement to the sound check, one which will deliver good, static baseline mixes that have the properties of predefined target mixes. A key contribution of this work is the inclusion of simplified acoustic-environmental system models, allowing the coupling behavior between multiple sources and receivers to be investigated. The coupling behavior highlights the difficulty faced by the engineer, particularly at small venues, where the coupling effects are more pronounced and where the available resources (in terms of equipment and time) are likely to be constrained.

Our approach treats the role of the audio engineer as a constrained, multi-objective optimization problem. This may be thought of as a sanitized view, devoid of the creative input of the engineer. The existence of the interactions between instruments and loudspeakers, and the requirement to prevent the onset of acoustic feedback, demand much of the engineer's attention. When considering setting the vocal level, there are an infinite number of ways in which the vocal signal can be spread among the loudspeakers, and finding the optimum seems much more suited to computation. We suggest that the system here would actually allow more creative input from the engineer by quickly dealing with the functional elements. We must always remember, however, that the computation is only effective if the objective is meaningful and well defined.

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